



THE PHYSICAL CONSTRAINTS OF EMBODIED ARTIFICIAL INTELLIGENCE

What the Physical Universe Demands of Any Proposed Motor Command

1. Time & Freshness

Is the evidence still true?

Governing Physics:

Information Age: $\Delta t = t_{\text{act}} - t_{\text{sens}}$
World Deadline: τ_{world}

Uncertainty: $\sigma(t) = \sigma_0 + v\Delta t$

Systems Metrics:

- Freshness Wall Δt_{wall}
- Tail Latency $(P_{99}, P_{99.9})$
- Latest-Frame Sampling
- Action Chunking (H steps)

Reaches:

All cognitive stages.

Failure Mode:

Stale-state collapse; open-loop drift during latency spikes.

2. Inertia & Stopping

Can the machine still stop?

Governing Physics:

Momentum: $\mathbf{p} = m\mathbf{v}$
Kinetic Energy: $E_k = \frac{1}{2}mv^2$

Braking Decel: $a_{\text{max}} \leq \mu g$

Systems Metrics:

- Stopping Envelope d_{stop}
- Linear Reaction: $v \cdot \Delta t$
- Quadratic Brake: $\frac{v^2}{2a_{\text{max}}}$
- Clearance Barrier: d_{clear}

Reaches:

Perception, Memory, Planning, Execution.

Failure Mode:

Kinetic overrun; obstacle collision during braking.

3. Actuation Limits

How fast may commands change?

Governing Physics:

Torque Rate: $\dot{\tau} = d\tau/dt$
Jerk: $\ddot{\mathbf{q}} = d^3\mathbf{q}/dt^3$

Shear: $\sigma = \tau/W\rho \leq \sigma_{\text{yield}}$

Systems Metrics:

- Torque Rate Clamp
- C^2 Jerk-Continuous Splines
- Backlash & Compliance
- Torque Saturation

Reaches:

Planning and Execution.

Failure Mode:

Gearbox tooth shear; resonance chatter.

4. Energy & Thermal

What does it cost to sustain?

Governing Physics:

Joule Heat: $C_{\text{th}}\dot{T} = I^2R - \frac{\Delta T}{R_{\text{th}}}$

Continuous: $I_{\text{cont}} = \sqrt{\Delta T_{\text{max}}/(RR_{\text{th}})}$

Junction: $T_j \rightarrow$ DVFS throttle

Systems Metrics:

- I^2t Accumulation Clamp
- Shared Power Rail Budget
- Thermal-State Provenance
- Extrinsic Calibration Drift

Reaches:

Mounts, Accelerators, Motors.

Failure Mode:

Winding burnout; silent clock throttle past deadline.

5. Silicon Determinism

Will the chip respond on time?

Governing Physics:

Crossbar Saturation: UMA DRAM
Ingestion Tax: DMA vs NPU

SRAM Mailbox: Lock-Free Ring

Systems Metrics:

- Pre-Inference Bus Tax
- Heterogeneous Coherency
- Zero-Heap Real-Time Rule
- Hardware Watchdog Leases

Reaches:

All compute substrates.

Failure Mode:

Memory bus starvation; non-deterministic heap pause.



THE CO-DESIGN INVARIANT: Every cognitive action proposed by software must satisfy all physical constraints before motor inverters are energized.